

NF W5 - Shell Items (LNK File) Analysis Report

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Abstract

LNK files, also known as Windows shortcut or shell link artifacts, are automatically created by Windows when a user opens files, folders, or certain applications. Although they are small background artifacts, they can preserve highly useful forensic metadata such as file names, file paths, working directories, and timestamp information that helps an investigator reconstruct user activity. In this lab, shell link evidence was generated through ordinary file interaction on a Windows system, extracted from the user's Recent folder with LECmd.exe, exported to CSV format, and reviewed in Timeline Explorer. The screenshots and output examined in this report demonstrate that LNK artifacts can identify recently accessed items, help isolate suspicious entries, and preserve meaningful evidence even when the original file or folder has been removed.

Introduction

The purpose of this assignment was to examine Shell Items, specifically LNK files, as a source of user-activity evidence in a Windows forensic investigation. In digital forensics, investigators rarely rely on a single artifact. Instead, they collect multiple traces left behind by the operating system and by ordinary user behavior. Windows artifacts are especially valuable because the operating system records many actions automatically in the background, often without the user realizing it. Those artifacts can help show what happened on a system, when it happened, and which files or folders may have been involved.

Among Windows artifacts, LNK files are useful because they are tied directly to user interaction with files and folders. When a person opens a document, browses into a folder, or works with a file stored on the desktop or in another location, Windows may generate or update a shell link in the user's Recent folder. Even if the original item is later deleted, the LNK file can remain behind as evidence that the item was previously accessed. In this lab, the uploaded screenshots and extracted CSV data show how those artifacts can be collected, reviewed, sorted, filtered, and interpreted to reconstruct a short sequence of activity involving a test folder and several text files.

Background Information

A LNK file is a Windows shell link file that stores information used to reference another object, usually a file or folder. In practical forensic terms, a LNK file is more than a simple shortcut icon. It can contain metadata about the linked item, including timestamps, path information, and working-directory values that help reveal prior user interaction. These artifacts are frequently examined in

desktop forensics because they often survive normal user cleanup and provide evidence that can be compared with other sources of system activity.

On Windows systems, user-generated LNK artifacts are commonly found in the Recent folder associated with a specific user profile. In this lab, the relevant location was the user path under AppData\Roaming\Microsoft\Windows\Recent, which is also visible in the extraction command shown in the screenshots. When a user opens a file or folder, Windows can create or update a corresponding LNK entry there. Because the artifact is stored separately from the original file, the record may remain even after the target file has been deleted. That persistence is one reason shell link artifacts are important in forensic analysis.

The metadata inside LNK files is useful because it can help answer investigative questions. Source timestamps describe the LNK artifact itself, while target timestamps describe the original file referenced by the link. Other fields, such as Relative Path and Working Directory, can show where the accessed item was located on the system. In many cases, these details help an investigator move beyond guesswork and toward a documented reconstruction of user behavior.

Tools and Methods

Two tools were central to this lab: LECmd.exe and Timeline Explorer. LECmd.exe is a command-line parser used to process LNK files from a chosen directory and export the decoded results into a readable format such as CSV. Timeline Explorer is a forensic review tool that opens CSV output and allows the analyst to sort, filter, and tag rows for interpretation. Together, these tools support a straightforward workflow: collect shell link artifacts from the Recent folder, export them, and then inspect the data visually in a structured table.

The test activity shown in the uploaded screenshots involved normal user interaction with a folder named Forensic Data and several text files inside it. The screenshots show a desktop folder being created and accessed, a text file being opened and reviewed in Notepad, and an item named Malicious Code.txt being deleted through the Windows context menu. After that activity, LECmd.exe was run against the Recent folder, and the exported CSV was opened inside Timeline Explorer for review. The resulting entries were then examined through sorting, filtering, timestamp comparison, and path analysis.

Procedure

The lab began with the creation and use of a test folder on the Windows desktop. Inside that folder, several text files were present, and at least one of them was opened in Notepad. Another item, Malicious Code.txt, was also present and later deleted. These normal actions were important because they generated the Windows activity necessary for LNK artifacts to be created in the Recent folder.

After the test activity was completed, LECmd.exe was executed from Command Prompt. The command shown in the screenshots targeted the user's Recent folder and directed the parsed output to the Desktop as a CSV file. The command window confirms that multiple LNK files were found and processed successfully. This step converted raw shell link artifacts into a format that could be examined more easily.

Next, Timeline Explorer was opened from the DFIR tools location and the generated CSV file was selected from the Desktop. Once loaded, the data appeared in a table view showing rows for individual LNK entries and columns for fields such as source timestamps, target timestamps, relative path, and working directory. From there, the data was reviewed using built-in features such as column sorting, text filtering, and tagging. Those features helped narrow the evidence to the entries most relevant to the simulated activity.

Finally, the relevant LNK rows were interpreted to determine what they revealed about user behavior. The screenshots show the analyst isolating suspicious content, viewing timestamp relationships, confirming the directory path of accessed files, and separating relevant entries from ordinary background system items. This final stage turned the extracted data into a forensic narrative instead of leaving it as a simple list of file names.

Figure-by-Figure Analysis

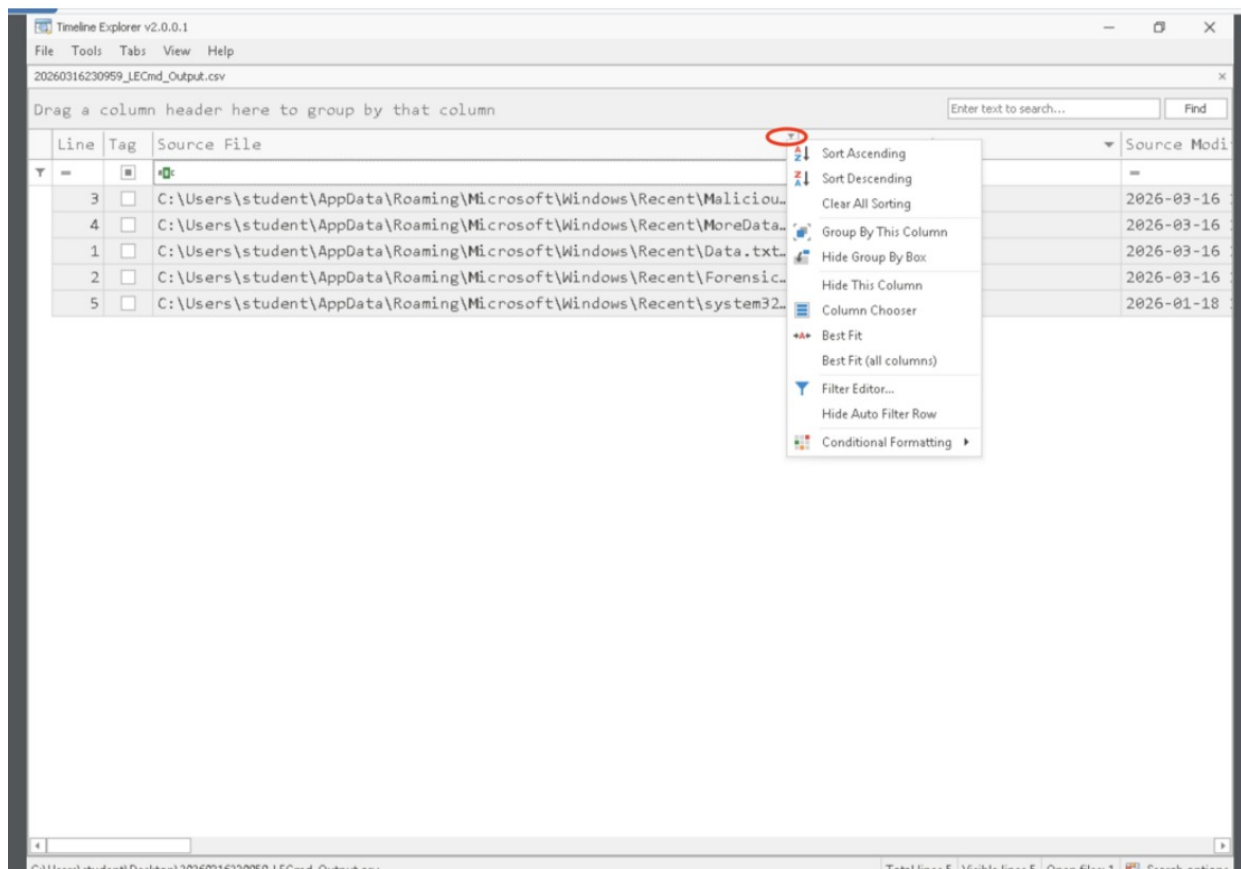


Figure 1. Timeline Explorer sort menu displayed from the Source File column.

This screenshot shows the context menu opened from a column header inside Timeline Explorer. The visible action is the use of sorting and grouping controls to organize the shell link data after the CSV was loaded. For a forensic investigator, this matters because chronological and alphabetical sorting are often the quickest ways to isolate the most relevant artifacts, especially when many routine system entries appear alongside the items tied to the user's activity.

Line	Tag	Source File	Source Cre...	Source M
1	☒	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Data.txt.lnk	2026-03-16...	2026-03-
2	☒	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Forensic Data.lnk	2026-03-16...	2026-03-
3	☒	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Malicious Code.txt.lnk	2026-03-16...	2026-03-
4	☒	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\MoreData.txt.lnk	2026-03-16...	2026-03-
5	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\system32.lnk	2026-01-18...	2026-01-

Figure 2. Source File entries displayed in Timeline Explorer after the CSV import.

This figure shows several LNK records listed in the Source File column after the export was loaded into Timeline Explorer. File names associated with the Recent folder are visible in table form, which allows the analyst to compare entries side by side and identify the records linked to the test activity. This matters because it demonstrates that shell link artifacts can be collected in a structured format and reviewed efficiently rather than examined one file at a time.

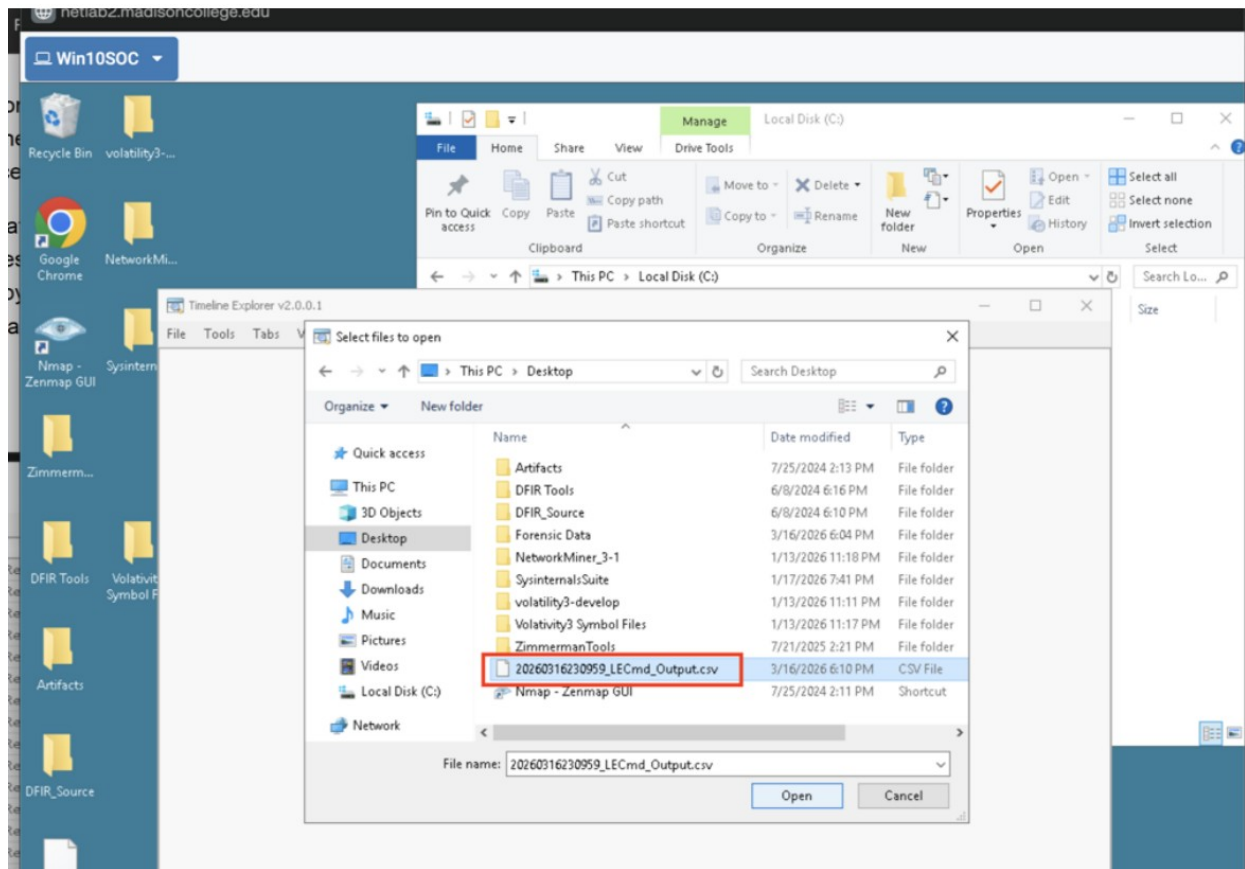


Figure 3. Open dialog used to select the exported LECmd CSV file.

This screenshot captures the step where the analyst browses to the Desktop and selects the CSV output file for review. It documents the transition from extraction to interpretation by showing how the exported evidence is opened in a forensic timeline viewer. That matters because a forensic report should explain not only what was found, but also how the artifact data was loaded into a tool for analysis.

Line	Tag	Source File	Source Cre...	Source M
1	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Data.txt.lnk	2026-03-16...	2026-03-
2	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Forensic Data.lnk	2026-03-16...	2026-03-
3	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Malicious Code.txt.lnk	2026-03-16...	2026-03-
4	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\MoreData.txt.lnk	2026-03-16...	2026-03-
5	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\system32.lnk	2026-01-18...	2026-01-

Figure 4. Initial Timeline Explorer table showing multiple shell link records.

This screenshot shows the LNK data displayed in a table after the CSV file was opened successfully. Several rows are visible at once, allowing the analyst to compare entries, identify likely evidence, and distinguish user-generated items from unrelated background entries. For a forensic investigator, this screen is important because it is where the artifact data becomes searchable, sortable, and ready for interpretation.

Sim	Relative Path	Working Directory
0	..\..\..\..\Desktop\Forensic Data\Data.txt	C:\Users\student\Desktop\Forensic Data
0	..\..\..\..\Desktop\Forensic Data	
0	..\..\..\..\Desktop\Forensic Data\Malicious Code.txt	C:\Users\student\Desktop\Forensic Data
16	..\..\..\..\Desktop\Forensic Data\MoreData.txt	C:\Users\student\Desktop\Forensic Data
8576	..\..\..\..\..\Windows\system32	

Figure 5. Relative path and working-directory information visible in the output table.

This figure highlights path-related columns in Timeline Explorer, including values that help connect a shell link to the original location of the target item. Path information is useful because it ties the artifact to a specific directory rather than only to a file name. In an investigation, that helps show where the user accessed the item and strengthens the reconstruction of the activity sequence.

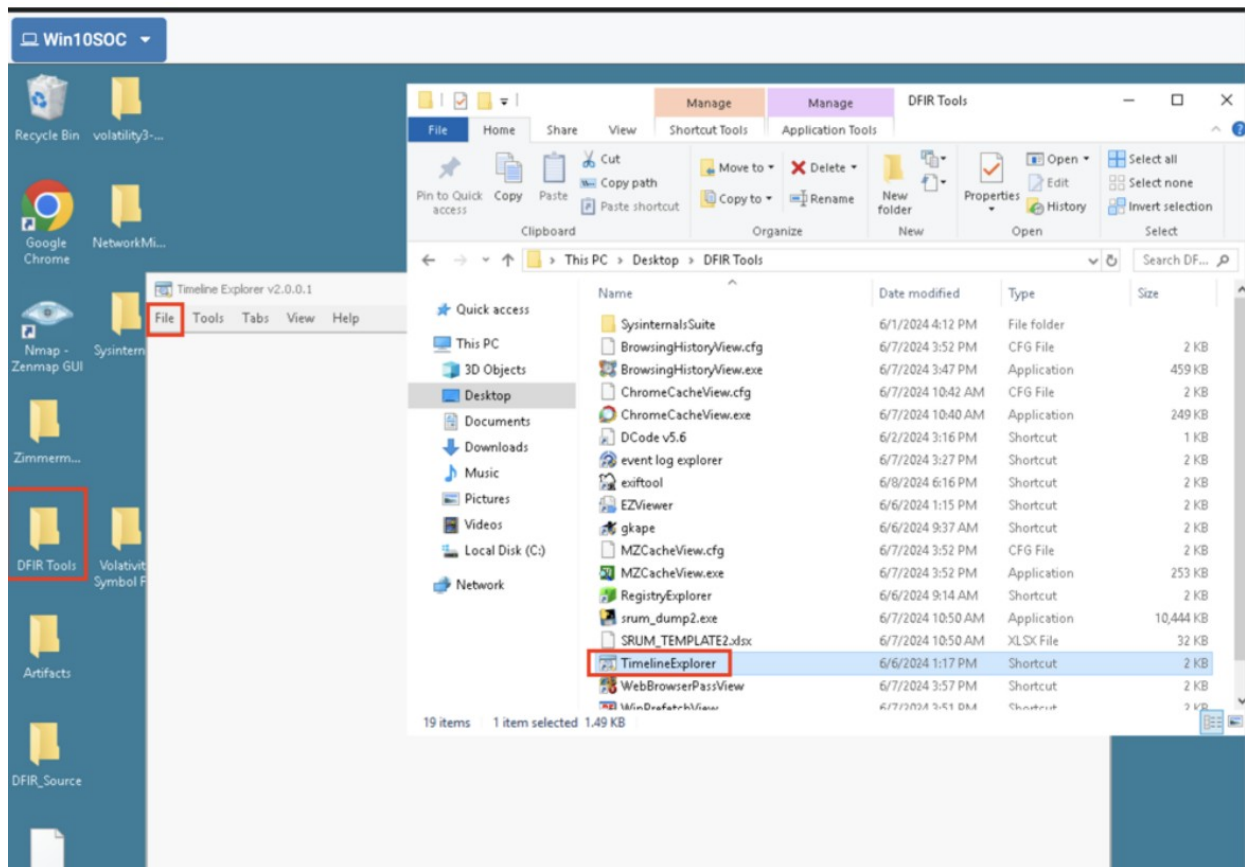


Figure 6. Timeline Explorer launched from the DFIR tools folder.

This screenshot shows the Timeline Explorer application being selected from the tools directory before the CSV evidence is reviewed. Although it is not evidence by itself, it documents the analysis workflow and helps explain how the shell link output was examined after extraction. That matters because forensic reporting should clearly describe the tools used and the order in which the analyst interacted with the data.

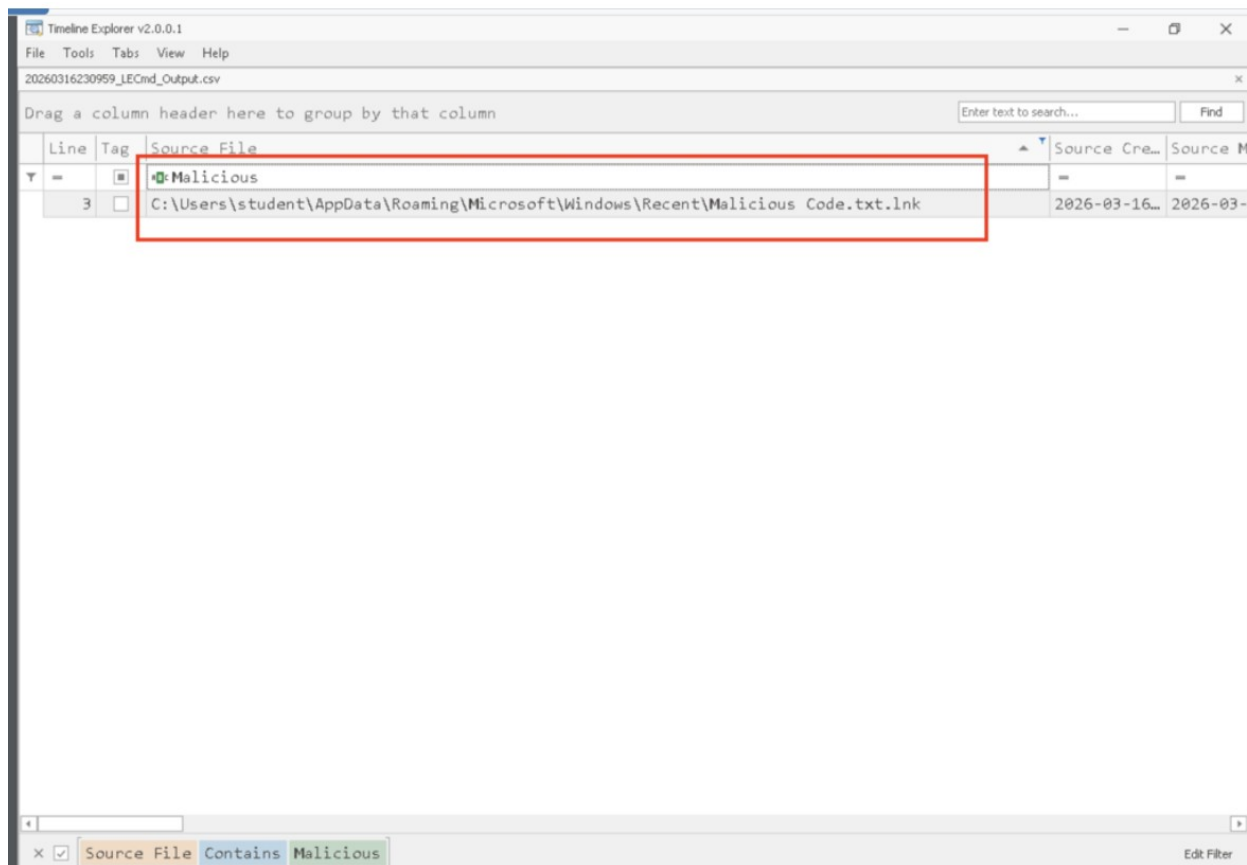


Figure 7. Filter applied for the keyword “Malicious” to isolate a single entry.

This screenshot shows the filter row being used to search for the word “Malicious” in the Source File column. The result is that Timeline Explorer narrows the table to the LNK record associated with Malicious Code.txt, making the suspicious item easy to isolate. This is significant because it demonstrates that even after the original file is removed, the shell link artifact can still be identified and reviewed quickly.

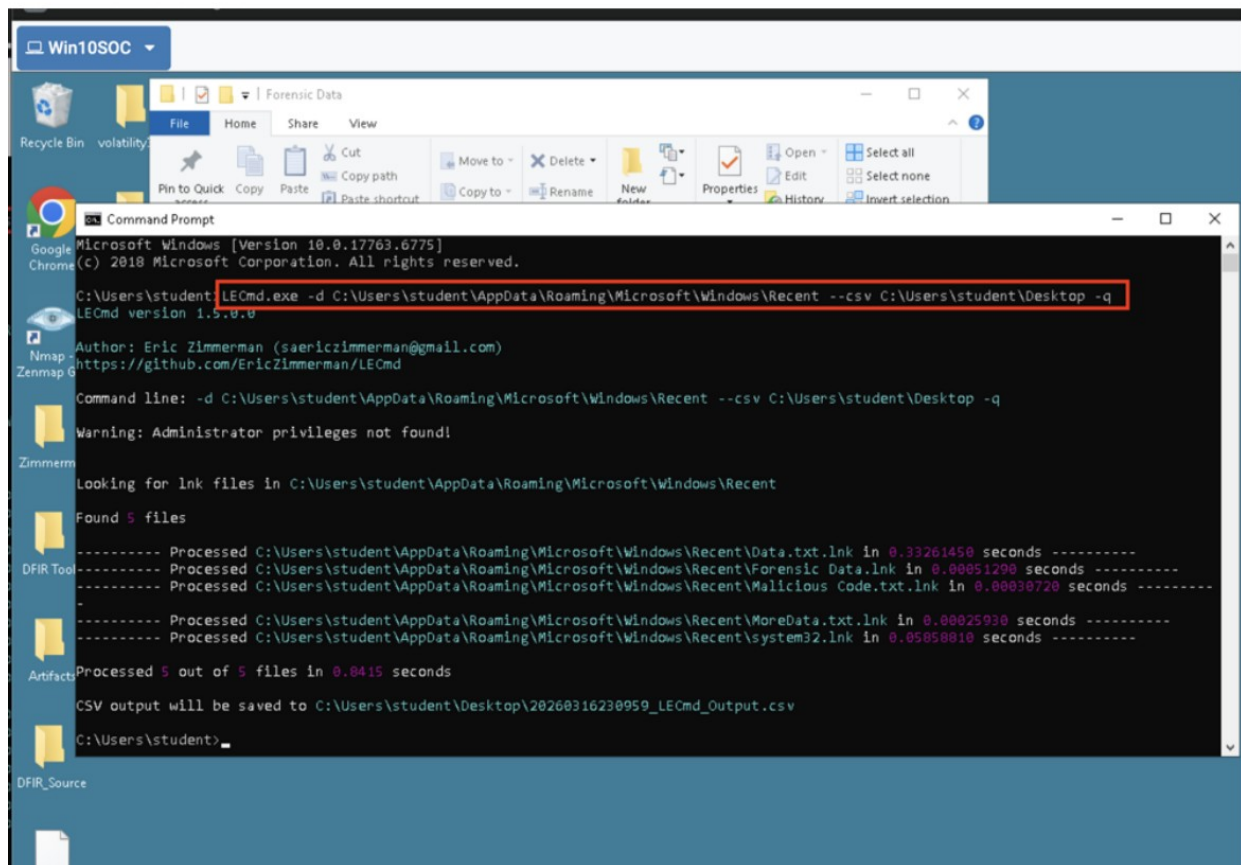


Figure 8. LECmd.exe command run against the Windows Recent folder.

This screenshot shows Command Prompt running LECmd.exe with the user's Recent folder as the source and a CSV file on the Desktop as the output. The command documents the extraction step and shows that multiple LNK files were parsed after the test activity took place. For a forensic investigator, this matters because it establishes how the evidence was acquired from a known artifact location rather than guessed from memory or visual inspection alone.

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

20260316230959_IECmd_Output.csv

Drag a column header here to group by that column

Enter text to search... Find

	Source Created	Source Modified	Source Accessed	Target Created	Target Modified	Target Accessed
▼	=	=	=	=	=	=
	2026-03-16 22:44:24	2026-03-16 22:44:24	2026-03-16 23:10:00	2026-03-16 22:44:06	2026-03-16 22:44:...	2026-03-16 22:44:06
	2026-03-16 22:43:31	2026-03-16 22:58:15	2026-03-16 23:10:00	2026-03-16 22:43:24	2026-03-16 22:48:...	2026-03-16 22:58:15
	2026-03-16 22:48:33	2026-03-16 22:48:33	2026-03-16 23:10:00	2026-03-16 22:48:12	2026-03-16 22:48:...	2026-03-16 22:48:12
	2026-03-16 22:44:50	2026-03-16 22:58:15	2026-03-16 23:10:00	2026-03-16 22:44:06	2026-03-16 22:57:...	2026-03-16 22:57:26
	2026-01-18 21:56:40	2026-01-18 21:56:40	2026-03-16 23:10:00	2018-09-15 06:09:26	2026-01-18 21:34:...	2026-01-18 21:56:40

Figure 9. Source and target timestamp columns displayed for comparison.

This figure shows timestamp columns such as Source Created, Source Modified, Source Accessed, Target Created, and Target Modified. These fields help distinguish between times associated with the shell link artifact and times associated with the linked file itself. In an investigation, timestamp comparison is important because it helps reconstruct when the artifact was created or updated and whether the target file appears to have been recently modified.

Timeline Explorer v2.0.0.1

File Tools Tabs View Help

20260316230959_LECmd_Output.csv

Drag a column header here to group by that column

Enter text to search... Find

Line	Tag	Source File	Source Cre...	Source M...
1	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Data.txt.lnk	2026-03-16...	2026-03-
2	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Forensic Data.lnk	2026-03-16...	2026-03-
3	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\Malicious Code.txt.lnk	2026-03-16...	2026-03-
4	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\MoreData.txt.lnk	2026-03-16...	2026-03-
5	☐	C:\Users\student\AppData\Roaming\Microsoft\Windows\Recent\system32.lnk	2026-01-18...	2026-01-

Figure 10. Timeline Explorer view sorted to prioritize the most relevant activity.

This screenshot shows the table arranged to emphasize the entries most useful to the analysis. Sorting helps surface recent or related records first and reduces the time needed to find the artifacts tied to the simulated user actions. This matters because a clear investigative method requires the analyst to move from a broad dataset toward a focused set of evidence that can be explained logically.

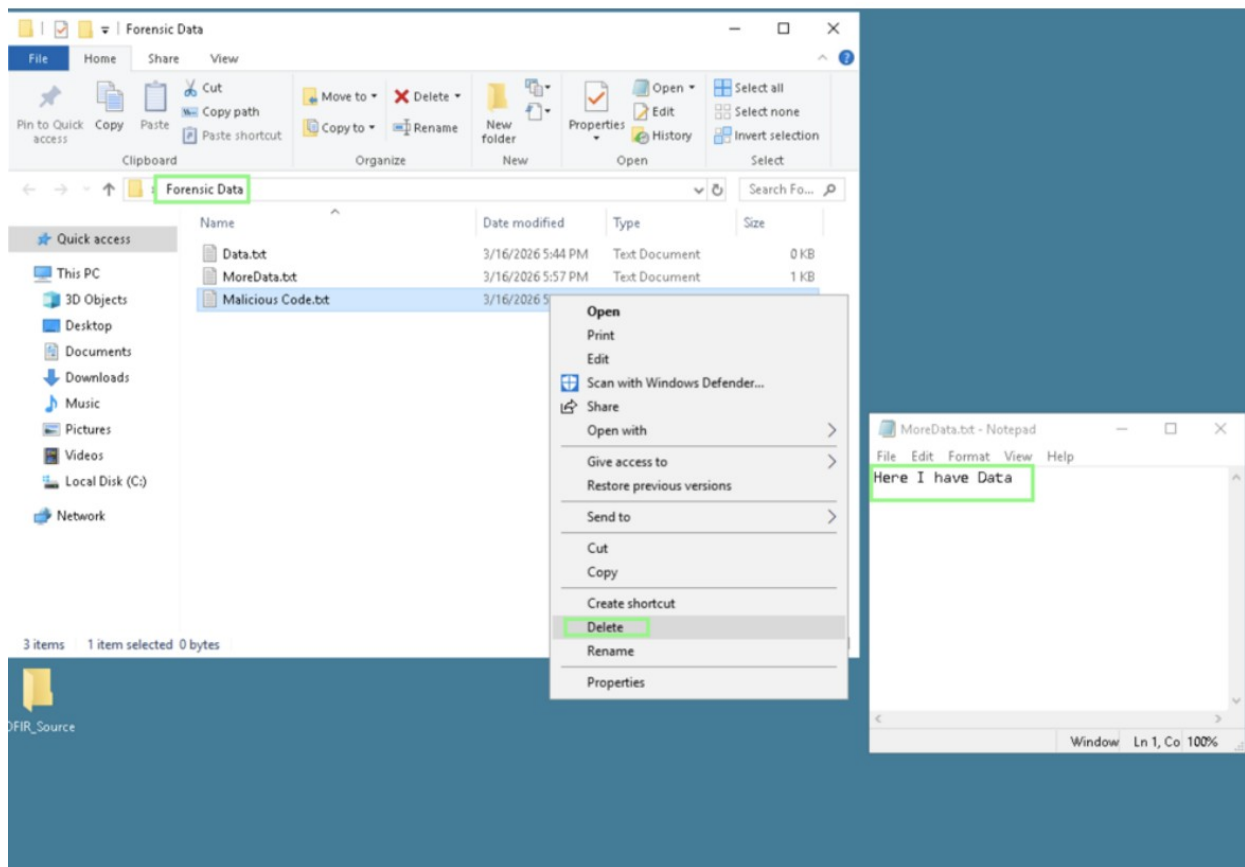


Figure 11. Deletion of Malicious Code.txt inside the Forensic Data folder while another file is open in Notepad.

This screenshot shows the Forensic Data folder on the desktop with several text files visible and a right-click menu over Malicious Code.txt with the Delete option selected. A Notepad window is also open, showing that another file from the same folder was being viewed during the activity. For a forensic investigator, this image is important because it captures the user behavior that later produced the shell link evidence: folder access, file interaction, and deletion of a suspicious item.

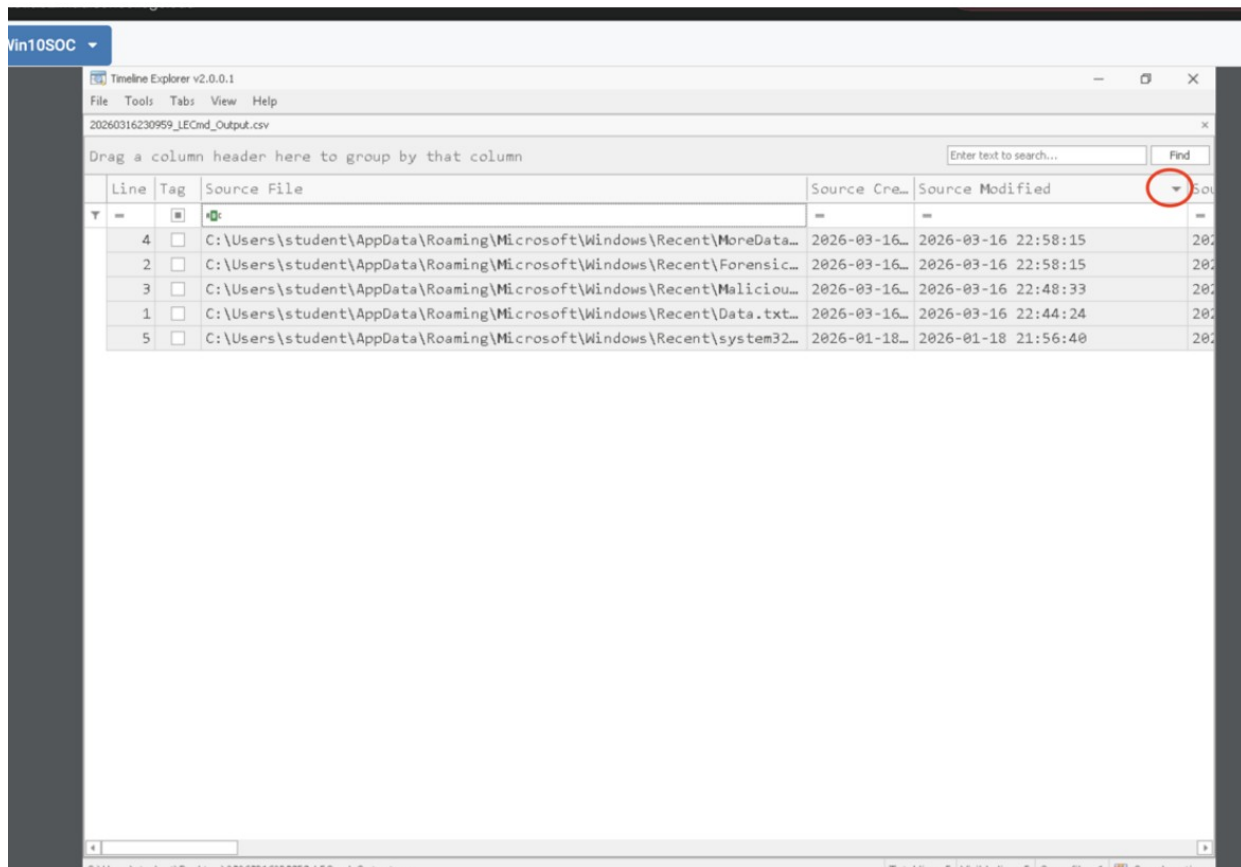


Figure 12. Additional Timeline Explorer review options displayed after the evidence load.

This screenshot shows another review screen inside Timeline Explorer with an emphasis on column-level controls after the CSV was loaded. These interface options support tasks such as sorting, narrowing visible fields, and focusing attention on the metadata most useful to the investigation. That matters because shell link analysis depends not only on collecting artifacts, but also on organizing and interpreting them in a disciplined way.

Data Interpretation

The Timeline Explorer output is most useful when the analyst understands what the major columns represent. Source Created refers to when the LNK artifact itself was first created. In practical terms, that often aligns with the first recorded interaction that caused Windows to generate that shell link. Source Modified reflects when the LNK was last updated, which can help indicate the most recent access represented by that artifact. Source Accessed refers to access involving the shell link file itself.

The Target Created and Target Modified fields relate to the original item referenced by the LNK rather than to the LNK artifact itself. These columns are useful because they help distinguish the shortcut metadata from the underlying file history. In this lab, the timestamp comparison is especially relevant for the files inside the Forensic Data folder because the activity included opening and interacting with text files before the artifact output was reviewed.

Relative Path and Working Directory help show where the target item was located when it was accessed. In the screenshots, the working-directory values connect the relevant items back to the

same desktop test folder. That consistency matters because it supports a clear interpretation that the files were accessed from one specific user-created location rather than from unrelated system paths.

Sorting, filtering, and tagging are what make the table actionable. Sorting places the most useful evidence near the top. Filtering isolates a keyword such as “Malicious” in seconds. Tagging marks the entries that belong to the event sequence of interest. Together, those functions turn a broad collection of LNK data into a focused evidentiary set that can be explained logically in a report.

Forensic Significance

The recovered LNK evidence reveals a clear pattern of user behavior. The screenshots show access to a folder named Forensic Data, interaction with text files inside that folder, and deletion of Malicious Code.txt. Later analysis in Timeline Explorer still shows an entry linked to that file, meaning the deletion of the original item did not remove the artifact of access. That is the central forensic lesson of this lab.

LNK artifacts are valuable because they can demonstrate that a file or folder was opened, not merely stored somewhere on disk. In an investigation, that distinction matters. A suspicious document found on a system is one thing; evidence that the user actually opened it is stronger because it connects the file to user activity. Although shell link evidence should be correlated with other artifacts in a full investigation, it remains a powerful source of activity reconstruction on its own.

The screenshots and findings support a simple reconstruction: the user accessed the Forensic Data folder, opened one or more text files, interacted with Malicious Code.txt, deleted that item, and then artifact review still recovered evidence of that interaction from the Recent folder. This sequence explains why shell link analysis remains valuable in digital forensic investigations, especially when the visible target file no longer exists.

Non-Technical Summary

In simple words, Windows quietly keeps little notes in the background when someone opens files or folders. Those notes are called LNK files. They can show what was opened, where it was located, and when the activity happened. Even if the original file is later deleted, the note about it may still remain.

That matters in an investigation because deleting something does not always erase the signs that it was used. In this lab, a file called Malicious Code.txt was deleted, but a record of it still appeared during the LNK review. For a reader with no forensic background, the main point is that Windows often remembers file activity even after the visible file is gone.

Conclusion

This lab proved that Windows automatically creates and preserves LNK artifacts as a normal byproduct of file and folder activity. By generating test activity, extracting shell link data from the Recent folder with LECmd.exe, and analyzing the results in Timeline Explorer, it was possible to recover evidence showing which items were accessed, where they were located, and how the activity could be organized into a readable sequence. Most importantly, the report demonstrated that deleted

items may still leave meaningful traces behind. The continued presence of the shell link entry associated with Malicious Code.txt shows why shell link analysis is such a useful forensic technique. Even when a target file no longer exists, LECmd.exe and Timeline Explorer allow investigators to recover and interpret the remaining artifact evidence in a way that helps reconstruct user activity on a Windows system.

References

Eric Zimmerman. LECmd. GitHub repository and DFIR tool documentation.

<https://github.com/EricZimmerman/LECmd>

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SANS Institute. Timeline Explorer. DFIR utility documentation. <https://www.sans.org/tools/timeline-explorer/>

Magnet Forensics. Forensic Analysis of LNK Files. Technical blog and training material. <https://www.magnetforensics.com/blog/forensic-analysis-of-lnk-files/>